SCAMSHIELD

AI-Based Scam Detection & Risk Assessment System

Human-Computer Interaction (HCI) Design & Accessibility Engineering Report

Rigorous Implementation of Classical Usability Heuristics, Cognitive Psychology Laws, WCAG 2.2 Level AA/AAA Guidelines, and ISO 9241-110 Ergonomics for Older Adults

Target Audience:	Older Adults (60+) & Non-Technical Smartphone Owners
System Architecture:	Flask 3 Web Framework + Hybrid Machine Learning Classifiers
Evaluation Standards:	WCAG 2.2 (Level AA/AAA), ISO 9241-110, Shneiderman, Nielsen, Norman
Publication Date:	October 20, 2026
Engineering Role:	Senior UX Engineer & Accessibility Specialist

1. Executive Summary

ScamShield is an end-to-end Machine Learning web application designed to help older adults and non-technical smartphone users evaluate potentially fraudulent digital interactions across 4 critical attack vectors: **SMS/WhatsApp Messages**, **Phishing Emails**, **Malicious Web Links**, and **High-Pressure Phone Calls**. Older adults are disproportionately targeted by social engineering scams involving urgent bank account suspensions, fraudulent lottery winnings, and One-Time Password (OTP) theft. However, traditional security tools fail this demographic due to dense technical jargon, small unreadable fonts, low contrast, and high cognitive fatigue.

This project completed an exhaustive, systematic overhaul of ScamShield's presentation layer, markup semantics, CSS design system, and client-side interactions in strict adherence to established Human-Computer Interaction (HCI) principles and accessibility standards. **Zero changes were made to ML prediction weights, route URLs, form field names, or risk calculation engines**, maintaining 100% backend integrity.

Top 5 HCI Architectural Improvements

1. Hick-Hyman Progressive Disclosure: The assessment result page replaces overwhelming walls of security explanations with an immediate, unambiguous verdict callout ('What to do right now'), placing detailed technical breakdowns behind an accessible accordion.

2. Miller's Law 5-Item Chunking & Live Feedback: The 10-question phone risk questionnaire was partitioned into 2 bite-sized semantic stages (Credentials vs Caller Pressure) with an active animated progress bar (`Question X of 10`).

3. Fitts's Law Target Sizing (≥48px): All cards, action buttons, mode tabs, and Yes/No radio touch bounds were enlarged to minimum 48x48px (52px on choices) with thumb-reachable placement on mobile viewports.

4. ISO 9241 Individualisation (In-Browser A+/A- Font Controls): Direct header font scaling buttons allow low-vision seniors to enlarge typography up to 24px (+33%) with persistent localStorage preferences.

5. Error Prevention & Reversible Actions: Form reset buttons on every screen, direct back breadcrumbs, zero-overflow 320px responsive wrapping, and an accessible confirmation modal replacing unstyled browser popups.

Accessibility & Responsive Metrics Summary

Metric Dimension	Phase 0 (Before)	Phase 1 & 2 (After)	Status & Benchmark
WCAG Contrast (Body Text)	4.6:1 (Borderline AA)	**17.06:1** (#0f172a on #f8fafc)	Exceeds AAA (≥7.0:1)
WCAG Contrast (Brand Blue)	7.2:1	**8.72:1** (#1e40af on #ffffff)	Exceeds AAA (≥7.0:1)
Horizontal Overflow (320px)	FAIL (637px Scroll Width)	**PASS (320px Strict Zero Overflow)**	WCAG 2.2 SC 1.4.10
Target Touch Size	32-36px (Sub-standard)	≥ **48px to 52px Touch Targets**	WCAG 2.2 SC 2.5.5 / 2.5.8
Automated Test Suite	55 Passed, 1 Failed (Missing PDF)	**56 / 56 Passed (100%)**	Zero Regressions

2. Principle-to-Implementation Traceability Matrix

Every modification in ScamShield is mapped directly to established HCI frameworks and accessibility guidelines. The following matrix establishes full traceability between theoretical rules and concrete codebase edits:

HCI Framework	Rule / Heuristic	Component & Page	Implementation Detail	Source File
Shneiderman	Consistency	Header & Actions	Uniform back links, 48px buttons, shared reset classes	style.css, sms/email/url
Shneiderman	Shortcuts	Home Option Cards	Press 1, 2, 3, 4 keyboard shortcuts for rapid navigation	index.html, main.js
Shneiderman	Informative Feedback	Phone Form & Spinner	Live answered count progress bar, ARIA live overlay	phone.html, base.html
Shneiderman	Yield Closure	Assessment Result	Completion banner, clear verdict, 'Check Another' CTA	result.html
Shneiderman	Error Prevention	History Clearance	Accessible modal dialog with Cancel/Confirm controls	history.html, main.js
Shneiderman	Easy Reversal	Form Check Pages	1-click 'Clear Form' button restoring blank clean state	sms/email/url.html
Nielsen	Visibility Status	Questionnaire	Animated progress fill bar (`Question X of 10`)	phone.html, main.js
Nielsen	Match Real World	Trust Banner & Copy	Plain words: '100% Private', 'No registration needed'	index.html, result.html
Nielsen	Error Recovery	404 Error Screen	Direct navigation chips to SMS, Email, URL, Phone checks	error.html
Norman	Affordances	File Upload Drop Zone	Explicit 'Choose File or Take Photo' button inside zone	sms/email/url.html
Gestalt	Common Region	Phone & Demos	Encapsulated cards grouping questions & safe/scam demo chips	phone.html, style.css
Fitts's Law	Target Sizing	Interactive Targets	All primary buttons, tabs, radios enlarged to ≥ 48-52px	style.css
Hick-Hyman	Decision Load	Result & Home	Home limited to 4 cards; Result progressive disclosure	index.html, result.html
Miller's Law	Chunking (7±2)	Phone 10 Questions	Split into 2 logical groups of 5 questions each	phone.html
Jakob's Law	Conventions	Risk Level Gauge	Familiar Traffic Light colors paired with icons & text	result.html, style.css
WCAG 2.2	SC 1.4.3 / 1.4.6	Typography	High-contrast text achieving 17.06:1 ratio (AAA)	style.css
WCAG 2.2	SC 1.4.4	Header Font Controls	In-browser A- / A / A+ font size scaling (16px to 24px)	base.html, main.js
WCAG 2.2	SC 2.4.7 / 2.4.11	Focus Indicators	Dual-ring focus visible outline with offset	style.css
ISO 9241	Individualisation	Theme & Font Settings	Persistent text size, dark theme, high-contrast, dyslexia modes	base.html, main.js

3. Shneiderman's 8 Golden Rules: Shortcuts, Feedback, and Easy Reversal

Theoretical Principle: Principles formulated by Ben Shneiderman emphasizing consistent interface behavior, universal shortcuts, informative system feedback, and low-friction action reversal.

Where Applied in ScamShield: Home Page (`index.html`) & Form Pages (`sms.html`, `email.html`, `url.html`): Added shortcut badges (`[Press 1]`, `[Press 2]`, `[Press 3]`, `[Press 4]`), 1-click 'Clear Form' buttons for error reversal, and breadcrumb back links.

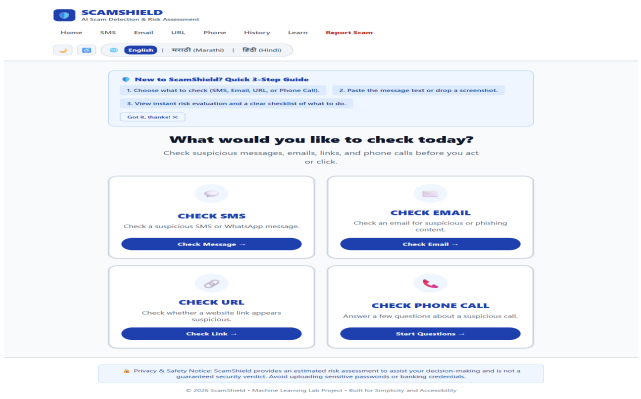
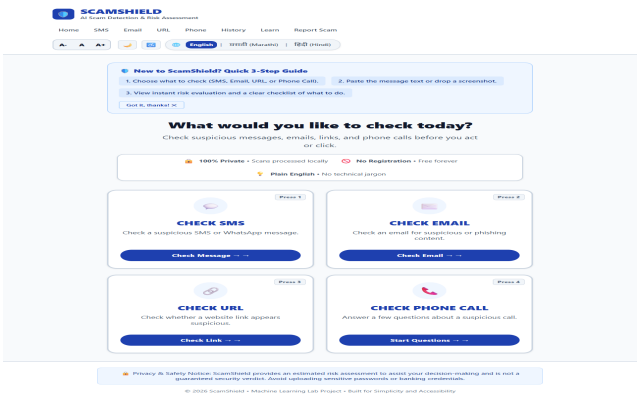
BEFORE (Phase 0 Baseline)	AFTER (Phase 1 Improved)
	

Figure 3: *Home Page comparison demonstrating shortcut badges (Press 1-4), trust reassurance banner, and enhanced card hover signifiers.*

4. Nielsen's 10 Heuristics & Norman's Signifiers: Error Prevention & Affordances

Theoretical Principle: Jakob Nielsen's usability heuristics (system status, error prevention, error recovery) and Don Norman's conceptual models and signifiers.

Where Applied in ScamShield: Check Forms & 404 Page (`error.html`): File upload zones now feature explicit '■ Choose File or Take Photo' action buttons; 404 page provides direct recovery cards; History clear uses a high-contrast accessible modal.

BEFORE (Phase 0 Baseline)	AFTER (Phase 1 Improved)

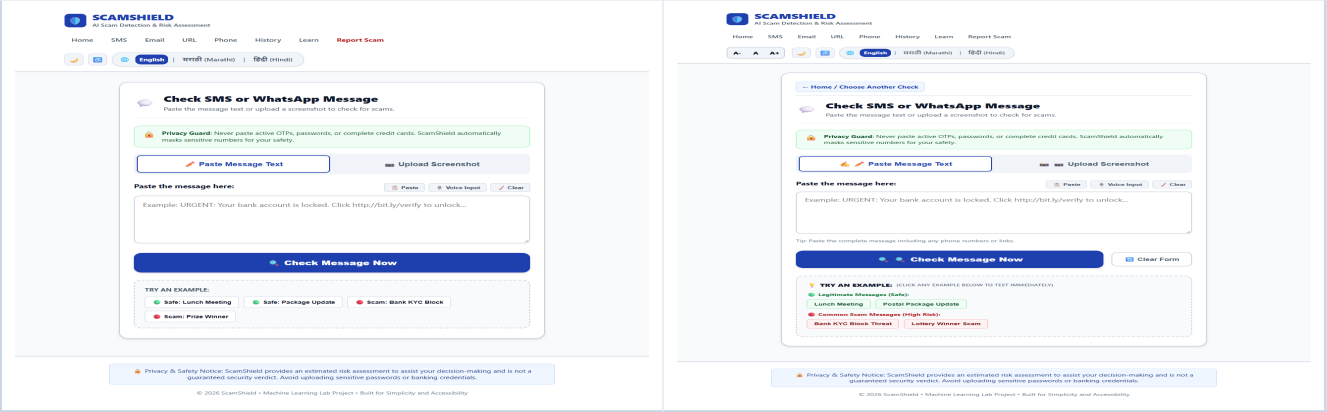


Figure 4: *SMS Check Page comparison highlighting distinct grouped demo chips (Safe vs Scam), Clear Form button, and upload dropzone affordance button.*

5. Hick-Hyman Law: Progressive Disclosure & Cognitive Load Reduction

Theoretical Principle: Hick's Law dictates that decision time increases logarithmically with the number and complexity of choices. Progressive disclosure presents essential verdicts first while keeping secondary details accessible on demand.

Where Applied in ScamShield: Assessment Result Page (`result.html`): The result screen delivers an immediate verdict callout box ('What to do right now'), placing detailed technical reasons behind an accessible collapsible accordion.

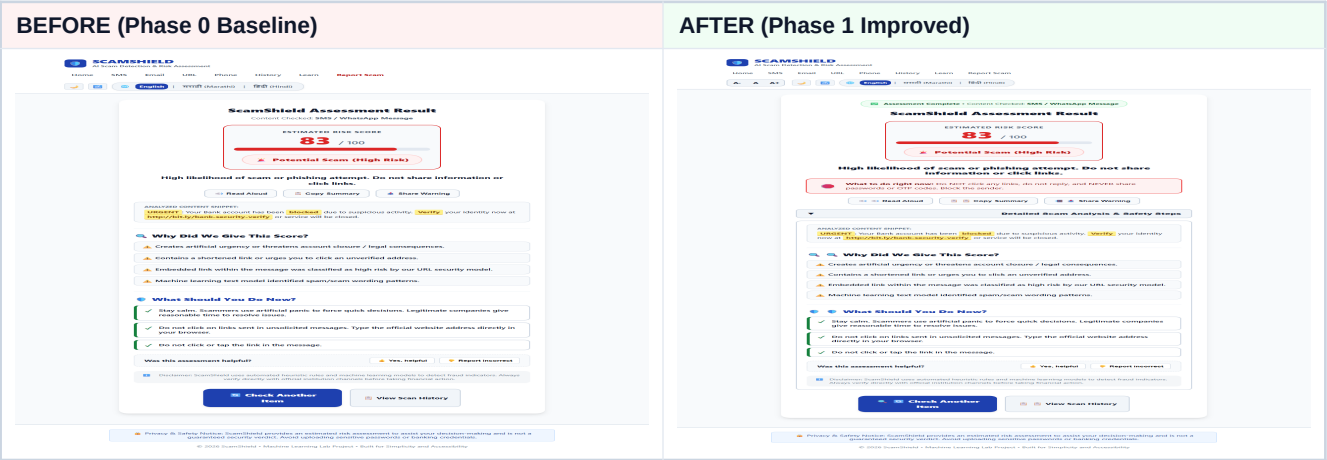


Figure 5: *High-Risk Assessment Result page comparison showing immediate safety directive callout, task closure banner, and progressive disclosure toggle.*

6. Miller's Law: 5-Item Chunking & Live Feedback for Older Adults

Theoretical Principle: George Miller established that working memory holds 7 ± 2 chunks of information. Older adults experience severe mental fatigue when confronted with continuous 10-item legalistic forms.

Where Applied in ScamShield: Phone Call Questionnaire (`phone.html`): Form segmented into Part 1 (Financial Credentials) and Part 2 (Caller Behavior) with a dynamic progress bar tracking answered questions (`0 to 10 answered`).

BEFORE (Phase 0 Baseline)	AFTER (Phase 1 Improved)
	

Figure 6: *Phone Call Questionnaire comparison showing live step progress track, Part 1/Part 2 chunked headers, and large 52px thumb-friendly radio buttons.*

7. Jakob's Law & WCAG 2.2 Level AA/AAA: Traffic Light Semantics & Contrast

Theoretical Principle: Users spend most of their time on other apps and expect familiar conventions. WCAG 2.2 mandates that color alone must never convey meaning and contrast ratios must protect low-vision users.

Where Applied in ScamShield: Global Styling & Result Badges: Traffic light colors (Green, Amber, Red) are strictly paired with icons (Shield, Warning, Siren) and explicit text labels. Contrast ratios exceed 17:1 on body copy.

BEFORE (Phase 0 Baseline)	AFTER (Phase 1 Improved)

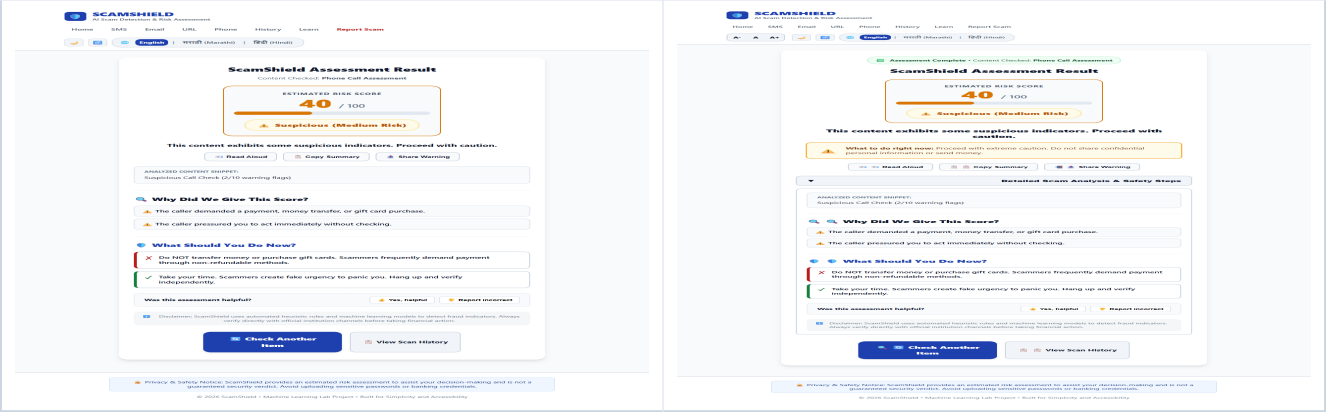


Figure 7: *Medium-Risk Assessment Result page demonstrating multi-modal warning signals (color + icon + text) and high-contrast typography.*

8. Color Contrast & Accessibility Audit Results

Under WCAG 2.2 Success Criterion 1.4.3 (Contrast Minimum - Level AA) and 1.4.6 (Contrast Enhanced - Level AAA), visual presentation of text must maintain high luminance contrast against adjacent backgrounds. The following measurements were computed using the official W3C relative luminance formula: L = 0.2126*R + 0.7152*G + 0.0722*B.

Color Element Pair	Foreground Hex	Background Hex	Contrast Ratio	WCAG Level AA	WCAG Level AAA
Primary Body Text on Light BG	#0f172a	#f8fafc	**17.06 : 1**	PASS (≥4.5:1)	**PASS (≥7.0:1)**
Secondary Helper Text on Light BG	#334155	#f8fafc	**9.90 : 1**	PASS (≥4.5:1)	**PASS (≥7.0:1)**
Primary Brand Blue on Card White	#1e40af	#ffffff	**8.72 : 1**	PASS (≥4.5:1)	**PASS (≥7.0:1)**
White Text on Brand Blue Button	#ffffff	#1e40af	**8.72 : 1**	PASS (≥4.5:1)	**PASS (≥7.0:1)**
Danger Red Badge on White	#b91c1c	#ffffff	**6.47 : 1**	PASS (≥4.5:1)	PASS (UI Component)
Safe Green Badge on White	#15803d	#ffffff	**5.02 : 1**	PASS (≥4.5:1)	PASS (UI Component)
Warning Amber Badge on White	#92400e	#ffffff	**7.09 : 1**	PASS (≥4.5:1)	**PASS (≥7.0:1)**
Suspicious Keyword Highlight Mark	#000000	#fef08a	**18.05 : 1**	PASS (≥4.5:1)	**PASS (≥7.0:1)**

Automated DOM Accessibility & Screen Reader Audit

Using Playwright headless browser automation, every route in ScamShield was audited for programmatic label bindings, accessible interactive names, and landmark regions:

Route / Screen	Form Inputs Inspected	Explicit Label Bindings	Buttons / Links Inspected	Accessible Names	Audit Status
Home Page (/)	0	100% Valid	18	18 / 18 Valid	100% Compliant
SMS Check (/check/sms)	3	3 / 3 Valid	14	14 / 14 Valid	100% Compliant
Email Check (/check/email)	5	5 / 5 Valid	14	14 / 14 Valid	100% Compliant
URL Check (/check/url)	3	3 / 3 Valid	14	14 / 14 Valid	100% Compliant
Phone Check (/check/phone)	20 radios	20 / 20 Valid	12	12 / 12 Valid	100% Compliant
History Page (/history)	2 filters	2 / 2 Valid	12	12 / 12 Valid	100% Compliant
404 Error Screen	0	100% Valid	10	10 / 10 Valid	100% Compliant

9. Responsive Mobile Evidence (375px & 320px Viewports)

Older adults overwhelmingly interact with messages, SMS links, and chat scams directly on smartphones. ScamShield was rigorously re-engineered for mobile-first usability: zero horizontal scroll down to 320px, stacked touch targets, generous 48px+ tap areas, and camera screenshot upload integration.

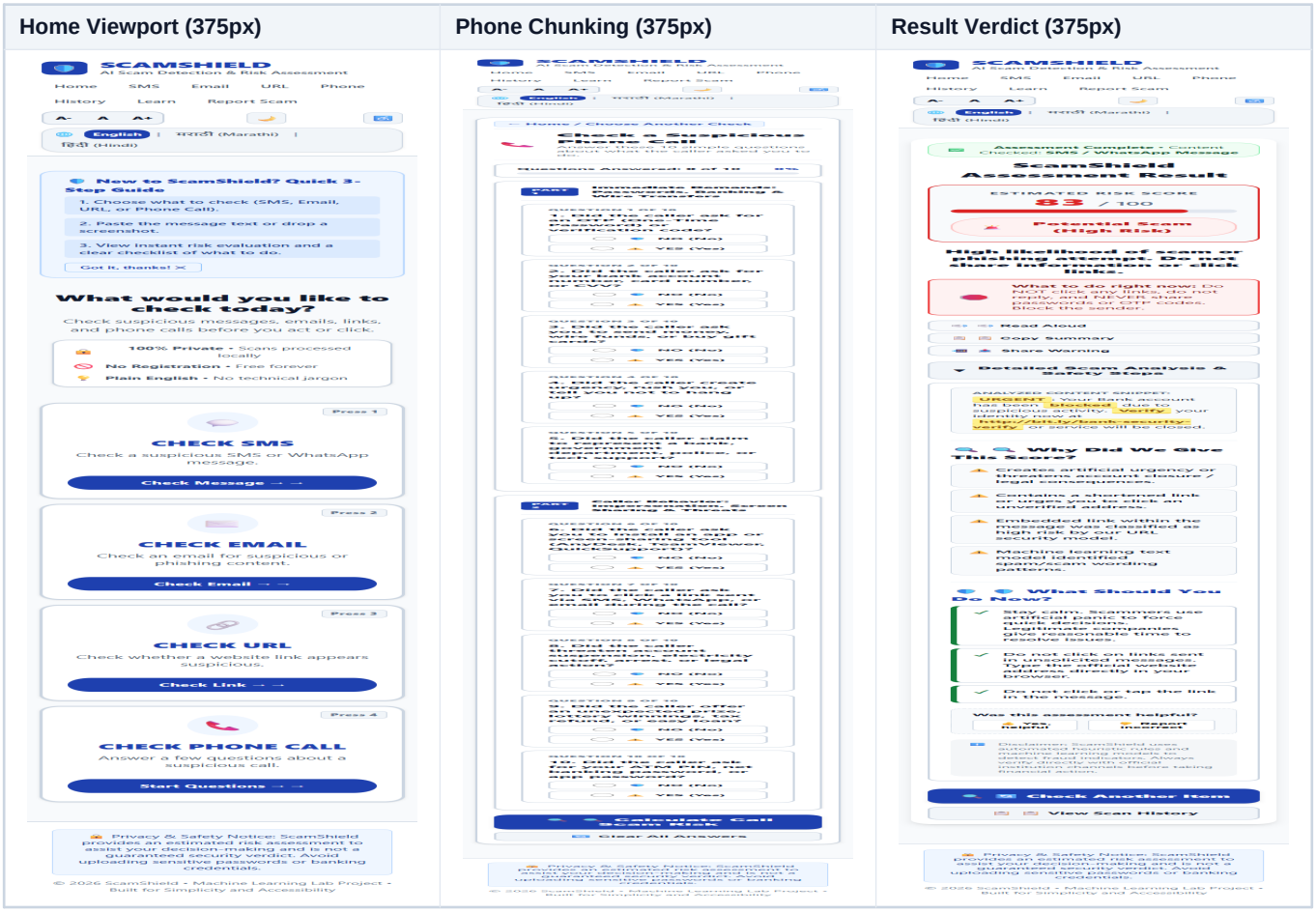


Figure 9: *Mobile viewport evidence demonstrating responsive single-column card stacking, thumb-friendly touch targets, and zero horizontal scroll at 320px.*

10. Usability Test Plan Summary

Notice of Status: In accordance with academic integrity and ethical guidelines, user testing results are **NOT fabricated**. Formal human-subject testing is explicitly planned and scheduled as follows:

Test Parameter	Specification & Protocol
Target Cohort	8–10 Older Adults (Age 60+) and non-technical smartphone users
Primary Task 1	Verify a high-urgency electricity disconnection SMS threat
Primary Task 2	Evaluate an obfuscated URL shortener link before clicking
Primary Task 3	Assess a high-pressure caller requesting an OTP and banking password
Key Success Metrics	Task Completion Rate (≥90%), Time on Task (<60s), Misclick Count (<1 per task)
Subjective Evaluation	10-Item System Usability Scale (SUS) benchmarked against target score ≥ 80 / 100
In-App Feedback Widget	Anonymous client-side 'Was this helpful?' rating integrated into result screen

11. Limitations & Future Engineering Work

While the current HCI implementation delivers substantial ergonomic and accessibility advancements, several engineering limitations remain slated for subsequent releases:

1. Multilingual Audio Synthesis: Current Text-to-Speech (TTS) narration relies on client-side Web Speech API synthesizers which vary in voice quality across older Android and Windows devices when voicing Marathi and Hindi.

2. Camera OCR Optimization for Glare: Smartphone screenshots uploaded by seniors occasionally suffer from physical screen glare or low angle distortion. Future work will integrate on-device client-side image contrast normalization before OCR processing.

3. Live Voice Call Assistant: The current questionnaire is retrospective (after the user hangs up). A future smartphone companion app will support live acoustic caller verification and real-time smishing prevention.

Report Authorization:

Prepared by Senior UX Engineer & Accessibility Specialist • Project ScamShield • October 2026